

Maximizing Medal Potential: A Strategic Approach to Selecting the US Women’s Artistic Gymnastics Olympic Team

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October 30, 2024

Abstract

Every four years, it comes to the exciting time for the USA Olympic Men’s and Women’s Artistic Gymnastics teams to compete at the international level, with Paris 2024 drawing near. The problem of selecting these teams is “complicated by the structure of the Olympic competition” and, to simplify our analysis, women’s team will be the primary focus of this report [2]. By analyzing a dataset from the 2024 UCSAS modeling competition, we examined key factors that potentially affect how the team performs in general [1]. Particularly, we employed basic exploratory data analysis to visualize elementary patterns. Then we computed a correlation matrix across event scores of individual athletes to spot potential relationships among them. Lastly, we conducted a paired t-test to identify any difference in domestic and international scoring, shedding light on significant discrepancy that might be associated with the event locations. Given the relatively small sample of data we possess, it becomes unlikely to test statistical significance for several patterns we observed, but other results can provide general guidance to follow when selecting the Olympic Team.

1 Introduction

The process of selecting the Olympic Team for Women’s Artistic Gymnastics by the US Olympic Committee is known to be an involved, meticulous, and strategic endeavor. Since we only have five available spots for every four years, the gymnasts to form our team should be carefully chosen in order to ensure overall expertise in the following four areas: Vault (VT), Uneven Bars (UB), Balanced Beam (BB), and Floor Exercise (FX).

Traditionally, the team final event will consist of three gymnasts on each event, which underscores the importance of having a well-rounded team. Given each gymnast’s unique strengths and weaknesses, we should carefully evaluate their potential contribution to ensure that the team can perform optimally and excel across all events. Hence, our objective here, as opposed to emphasizing individual talents, is to craft a cohesive team that maximizes the overall score and medal potential via statistical methods. At the end of this report, we provide important considerations that the US Olympic Committee should account for in this selection process.

To answer this prompt better, we break the objective and formulate three hypotheses detailed in Section 4, drawing insights from our exploratory data analysis. These hypotheses transition from broad to specific, guiding our investigation towards more focused inquiries. The rest of this report will be structured as follows: in Section 2 & 3, we summarize the dataset in use and how we cleaned it to ensure optimal representation. In Section 5, we discuss relevant data analysis and the contribution to answering these hypotheses. In Section 6, we reflect on limitations of this approach and shed light on future directions to work on. Lastly, Section 7 will be a conclusion that recapitulates our findings.

2 Dataset

Our dataset was provided by the 2024 UCSAS modeling competition, capturing major domestic and international competitions starting after the Tokyo 2020 Olympics and leading up to the Paris 2024 Olympics [1]. One single, complete observation in our data consists of the gymnast’s first name, last name, gender, home country, competition name, competition date, competition location, apparatus, rank, difficulty score, execution score, penalty, and overall score.

There are 1625 unique gymnasts represented, 2 genders reported by the gymnastics federation, 107 countries represented, 39 competitions, 6 different rounds, 30 major locations of competition, and 11 apparatuses (technically 9 since VT1 and VT2 are considered VT). The last few unique values are not of importance.

Missingness is very important in the dataset and is intentionally present. It represents a gymnast not competing all of the events in a single round of competition. For example, if a gymnast is a specialist (trains and competes on a select number of events), then they would show NA for every other event in every round of competition. If a gymnast has no NA values in a qualification round but several NA values in a final round, it means that they did not advance to finals in every event.

3 Data Preprocessing

We conducted some preliminary cleaning on this dataset. First of all, since we only consider women's team in this project, we dropped all of the observations of the male athletes by filtering on the gender column. Next we pivoted wider this dataset by isolating the four competing areas of the women's team in new columns, namely Vault (VT), Uneven Bars (UB), Balanced Beam (BB), and Floor Exercise (FX), each with the associated scores.

As we are interested in studying the difference between domestic and international competitions, it is crucial to obtain a boolean column that identifies domestic ones. To achieve this, we utilized regular expressions to find matches between the event location and the team's nationality. Several subtle nuances were manually taken care of: for instance, England and United Kingdom are treated as the same location, and there are several nonexistent IOC country codes in the dataset. Since they only comprise an extremely small proportion of the data, we dropped them and assumed no significant consequence in doing so. The majority of the following analysis was done on the processed data.

4 Hypotheses and Approach

To tackle our overarching question of how should the US Olympic Committee choose the strongest gymnastics team, we chose to analyze three different facets of competition. These three questions we explored include:

1. **Which events contribute most to a team score?**

For our first question, we will use exploratory data analysis to identify patterns in a small sample of team finals data.

2. **Is there a relationship between the different event scores? Are some events more strongly correlated than others?**

For our second question, we will analyze the correlation matrix between event scores of individual athletes.

3. **Is there a difference between domestic and international scoring?**

For our third question, we will analyze a paired t-test of event scores of individual athletes who competed both domestically and internationally.

5 Results

In this section, we summarize key intuitions from some exploratory data analysis and formal use of statistical methods to consolidate these ideas. Two selected graphs from EDA help us glean information about the relationship between domestic competitions and international ones.

In [Figure 1](#) and [Figure 2](#), the boxplot and frequency distribution provide us with some hint on the subtle differences between the two types of competitions. While we can see much similarity across these plots, they seem to display slight discrepancy. This observation inspires us to perform a paired t-test to examine the potential existence of statistical significance for hypothesis 3.

For hypotheses 1 and 2, we will directly use a summary table and correlation matrix, as they are more straightforward.

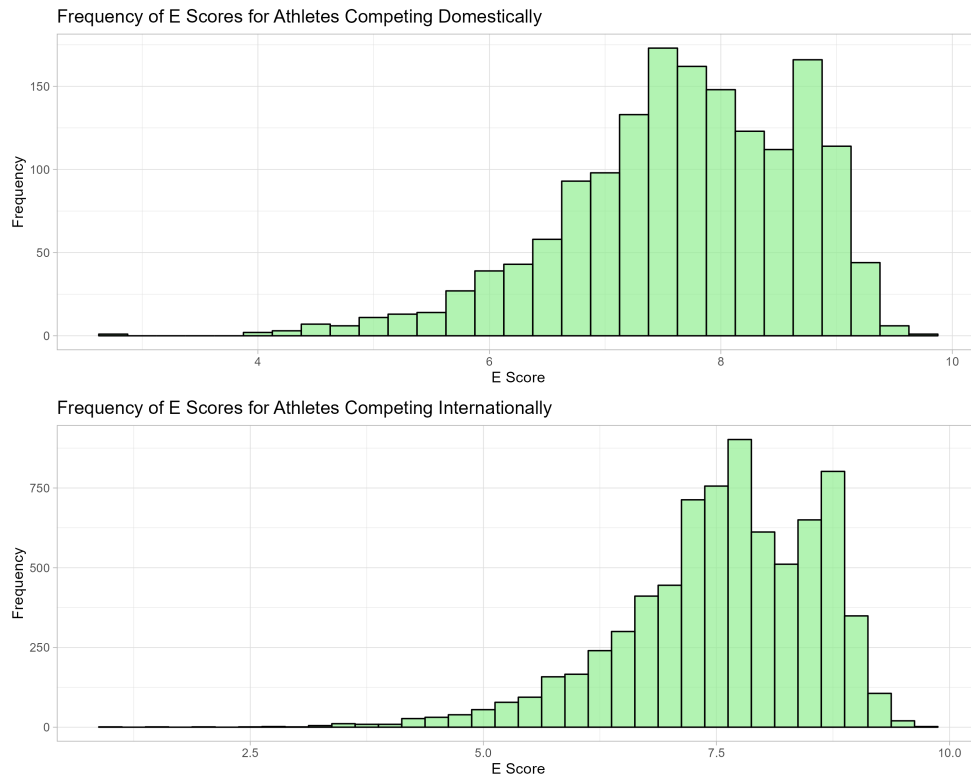


Figure 1: Frequency Distribution of Domestic vs. International E Scores

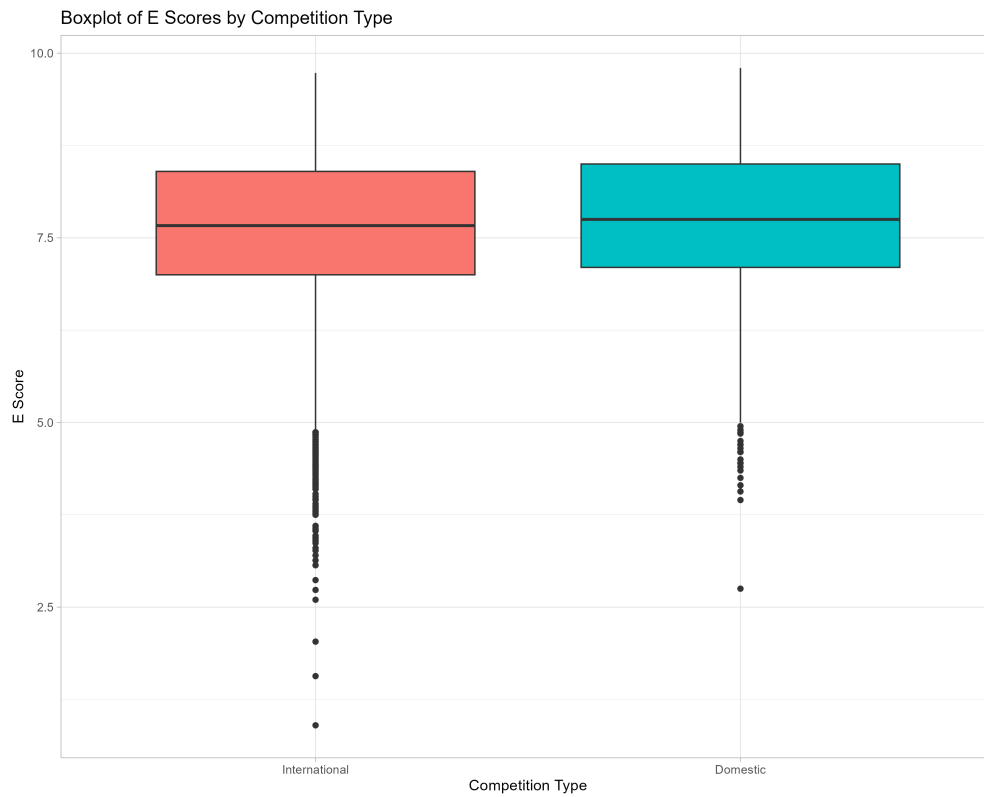


Figure 2: Boxplot of Domestic vs. International E Scores

5.1 Which events contribute most to a team score?

To effectively illustrate the contribution of each event to the team score, a table of the rankings across events and the final ranking provides a concise and clear presentation. There are eight teams in total. Rank 1 is first place while rank 8 is last place. Histograms of the team score distributions are provided in addition.

Competition	Country	Vault	Uneven Bars	Balance Beam	Floor	Total
2022 51st FIG Artistic Gymnastics World Championships	USA	1	1	3	1	1
2022 Senior European Championships	ITA	1	1	1	1	1
2023 52nd FIG Artistic Gymnastics World Championships	USA	1	1	6	2	1
2023 Artistic Gymnastics Senior Pan American Champion	USA	1	1	1	1	1

Table 1: Observed patterns in ranking for Team Finals - Winners

Competition	Country	Vault	Uneven Bars	Balance Beam	Floor	Total
2022 51st FIG Artistic Gymnastics World Championships	FRA	7	6	8	7	8
2022 Senior European Championships	ESP	8	8	4	6	8
2023 52nd FIG Artistic Gymnastics World Championships	JPN	7	8	3	8	8
2023 Artistic Gymnastics Senior Pan American Champion	PUR	8	6	8	8	8

Table 2: Observed patterns in ranking for Team Finals - Losers

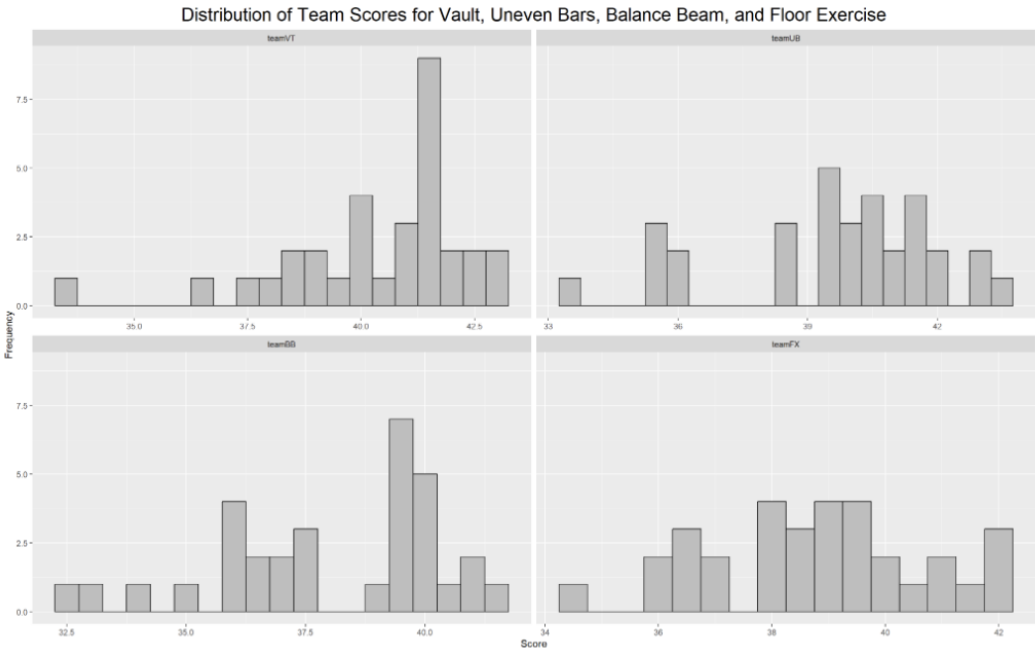


Figure 3: Distribution of Team Scores

As we can see in [Table 1](#), [Table 2](#), and [Figure 3](#), our initial pattern supports the idea that VT and UB are the most important events that contribute to a high final score, while BB has less of an impact on ranking first. A first place in Vaults and Uneven Bars is usually accompanied with the first place in total; on the other hand, placing high in Balance Bars does not seem to have an effect on the total ranking. However, our sample size is small, meaning that it only provides superficial intuition, and we will need to investigate other facets of our data in order to draw proper conclusions.

5.2 Is there a relationship between the different event scores? Are some events more strongly correlated than others?

To investigate whether there is a relationship between different events, we first pivoted our dataset wider such that each row represented a gymnast's performance in a certain round of competition for a certain competition. Thus, there were columns for each event: VT, UB, BB, and FX. We computed a correlation matrix as follows:

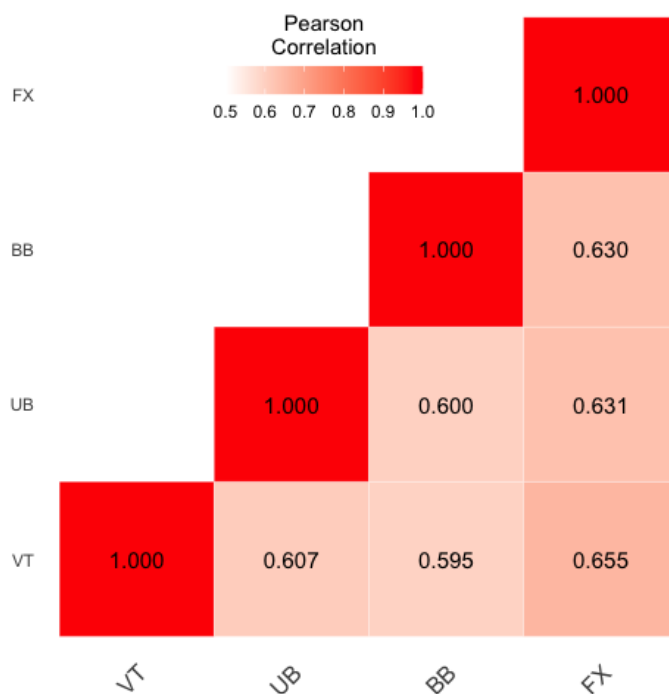


Figure 4: Heatmap, Correlation Matrix for Events

As we can see in [Figure 4](#), FX and VT are the two most strongly correlated events. VT and BB are the two least correlated events, and FX appears to be very correlated with every other event. This analysis sheds light on the interplay between different event scores and leads to insights for strategic team composition.

5.3 Is there a difference between domestic and international scoring?

To answer this question, we looked at the performance of U.S. athletes who had competed in both international and domestic competition. For each athlete, we computed their average E score at international competitions, their average E score at domestic competitions, then took the difference per a Paired-T procedure. Below is a graph displaying the differences:

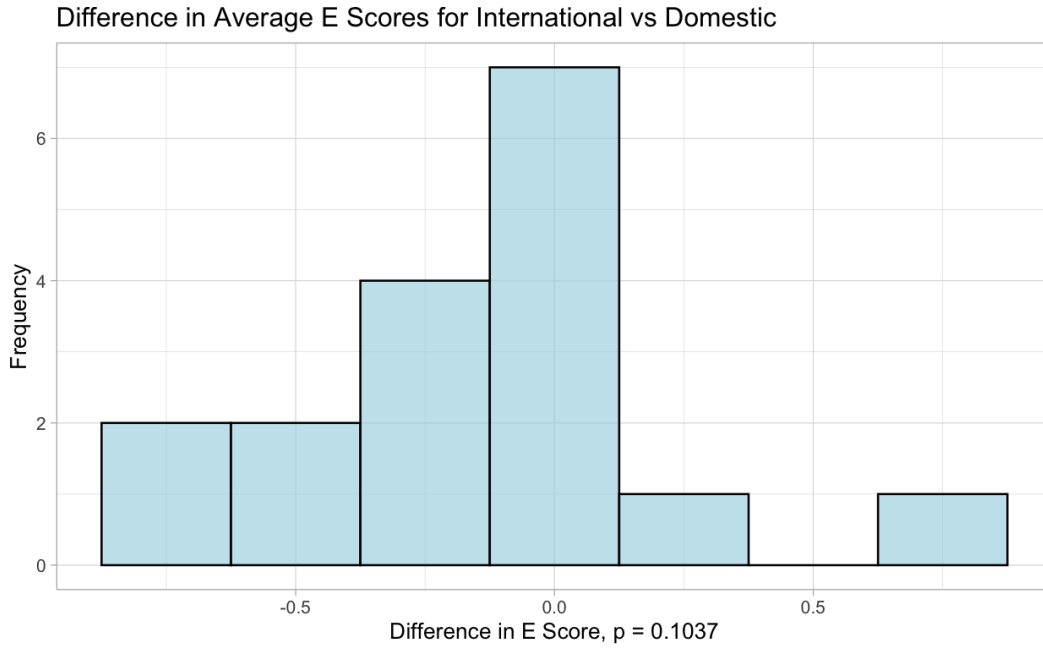


Figure 5: Frequency of Difference in E Scores, with p-value

As the p-value is not significant in [Figure 5](#), we lack sufficient evidence to conclude that there are significant discrepancies between domestic and international execution scoring. However, we furthered our analysis by breaking it down per event and running the same analysis.

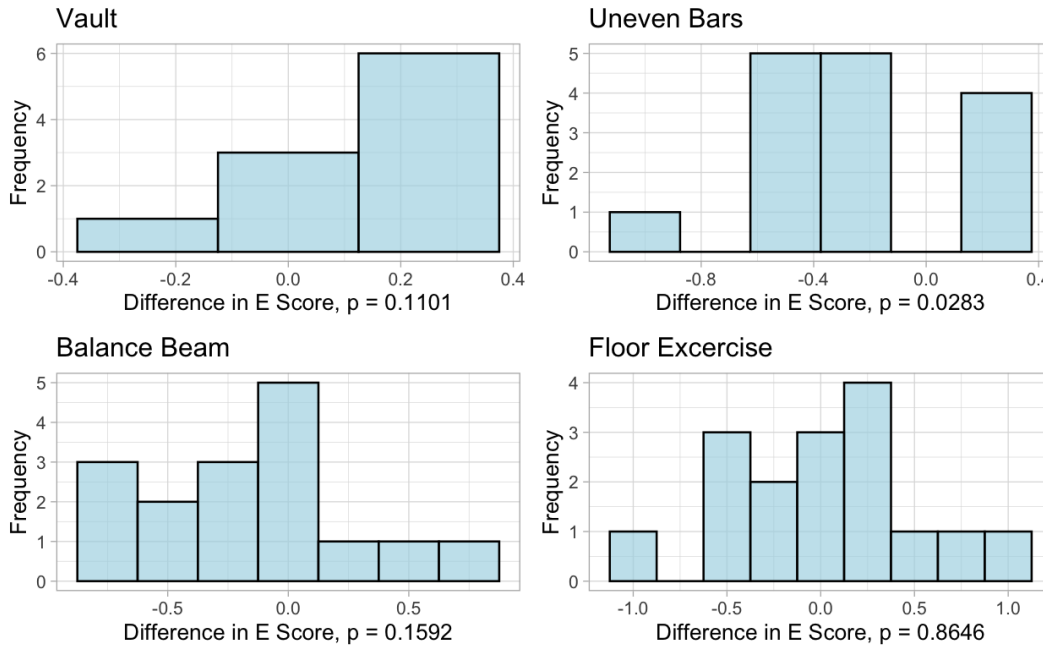


Figure 6: Frequency of Difference in E Scores for Individual Events, with p-value

Uneven Bars comes back as statistically significant in [Figure 6](#), suggesting that there is a difference between domestic and international scoring of U.S. athlete's uneven bars execution scores. This further emphasizes the importance of high-scoring uneven bars workers as those may be needed to overcome the international-domestic discrepancy.

6 Discussion and Limitations

As our analysis provides some insights about the selection process, our advice to the US Olympic Committee would be to prioritize strong Vaulters and Uneven Bars workers since they tend to contribute to a high final score. Since floor is heavily correlated to both events based on the observation, a good floor score will likely be taken care of as well. The Committee should select Balance Beam workers last, as they are not as correlated or as impactful for a team score in general.

Our results also help clear the misunderstanding that there might be significantly different result across domestic and international competitions. It turns out that the paired t-test for this dataset shows little significant discrepancy between the two, and the Committee should not worry too much about a so-called “international look,” which does not seem to correlate with team success.

One of the notable limitations associated with our approach is that it focuses on deriving relationships but lacks predictive power, as there was no model fitting or relevant regression techniques. One possible direction for future work is to harness the insights we generalize from these observations and adopt them in a model to predict a quantifiable aspect of team performance, for instance, the number of medals won by a team. We claim that much external data source would be available for this, but the modeling process can get tricky and tedious, as there are a number of possible predictors, many of which can have high order interaction effects. However, once accomplished, it can add power to our analysis and provide more solid basis for the choices made.

Additionally, a major assumption for the work is that the USA Olympic Men’s and Women’s Artistic Gymnastics teams are independent from each other. However, there can be implicit dependence between the two genders. For instance, coaching staff, training facilities, and overall program resources may be shared between the men’s and women’s teams, potentially impacting their performances in similar ways. Moreover, the public perception and media coverage of one team’s success or failure could influence the other team’s morale and confidence. Therefore, we might actually benefit from learning about the men’s team to explore the women’s team better. This provides an avenue for future, that is to investigate interactions between the two genders, which can help us learn both separately.

7 Conclusion

In conclusion, for this project, we aimed to strategically select the US Women’s Artistic Gymnastics Olympic Team for maximum medal potential, focusing on Vault (VT) and Uneven Bars (UB) due to their significant impact on final scores. Through exploratory data analysis and statistical testing, we identified that these events, particularly VT and UB, are crucial for team success. Our findings also highlight a statistical discrepancy in international versus domestic scoring for the Uneven Bars, suggesting the need for strong performers in this apparatus to bridge the scoring gap. However, the general discrepancy in the two types of competitions shows no statistical significance.

References

- [1] UCSAS. *Gym2024data*. URL: <https://github.com/ucsas/gym2024data/tree/main/cleandata>.
- [2] UCSAS. *UCSAS 2024 USOPC DATA CHALLENGE*. URL: <https://statds.org/events/ucsas2024/challenge.html>.